

An Exact Manual Shuffle

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Abstract

We give a procedure for shuffling a deck of cards by hand, directed by a die or coins, whose output is *exactly* uniformly distributed over all $n!$ orderings—not approximately, not asymptotically, but with zero error in every metric, conditional only on fair independent draws and one discipline: piles are collected in an order fixed in advance. The procedure is a single repeating structure that fits on a card table: deal the deck into a row of bins; collect the bins in predetermined order; any bin holding more than one card is dealt into the next row first, by the same rule. It requires no skill, no counting, never shows a card face, terminates almost surely in a few rows, and a bystander who observes everything but the card faces—including the random running time—learns nothing about the result. Run on a computer, the same rule turns a bare fair-bit generator into an exactly uniform permutation with output-independent timing. A 52-card deck costs about 121 card placements using a die and a coin in a twelve-slot box lid; a purpose-built 36-bin stand with single-roll resolution of two- and three-card piles reduces this to about 73 throws and 55 placements, with most runs never recursing at all. The procedure physically realizes the strong uniform time of Aldous and Diaconis; the proof is a paragraph.

1 The procedure

Fix a radix d and a randomizer providing independent uniform draws from $\{1, \dots, d\}$ per card:

$d = 6$:	one die	(six bins)
$d = 8$:	three coins read as binary	(eight bins)
$d = 12$:	one die and one coin	(twelve bins)

Lay out one *row* of d bin positions, numbered by draw value. The procedure is one rule, applied recursively:

To shuffle a pile: deal its cards one at a time, each to the bin its draw names, all face down. Then collect the bins in a fixed, predetermined order (say right to left): an empty bin contributes nothing; a bin holding one card goes onto the output; a bin holding two or more cards is first shuffled by this same rule, dealt into the next row below, and its result then goes onto the output.

*Procedure and layout by W. MacEvoy; analysis, simulations, and drafting assistance by Claude (Anthropic).

Shuffling the deck means applying the rule to the whole deck in row one. Each level of recursion is a fresh row of the same d bins, so the table layout is a small stack of identical rows used top to bottom; a pair of cards that keeps colliding simply walks one more row down, and separates there with probability $1 - 1/d$. Seven rows suffice outside a fraction $\binom{n}{2}/d^6$ of runs (about 3% for a 52-card deck at $d = 6$, 0.04% at $d = 12$); when the rare deeper collision occurs, reuse a cleared row.

2 The theorem

Theorem 1. *Suppose every draw is independent and uniform on $\{1, \dots, d\}$, and every collection order is fixed before the corresponding deal. Then the output of the procedure is exactly uniformly distributed on the $n!$ orderings of the deck. Moreover it is exactly uniform conditional on the entire observable transcript—every bin size in every row—so an observer who sees everything except card faces learns nothing about the result.*

Proof. Induction on the pile size n ; sizes 0 and 1 are trivial. Condition on the size vector $(b_1, \dots, b_d)$ of the top-level deal. Given sizes, the sets of cards in the bins are uniform over all partitions with those sizes, because the draws are i.i.d. uniform; and each bin’s internal order is uniform by the induction hypothesis, independently across bins. Concatenating in any order fixed independently of the draws therefore yields a uniformly random ordering of the pile, conditional on the size vector. Since this holds for every size vector, it holds unconditionally; and applying the same argument at every level of the recursion gives uniformity conditional on the full size transcript. Termination is almost sure: any two cards still sharing a bin separate at the next row with probability $1 - 1/d$. $\square$

Corollary 2. *The output permutation is statistically independent of the running time and of every function of the size transcript: duration, number of placements, number of re-deals, and rows reached may be disclosed without any loss of key material. (Uniform conditional on every transcript value means the joint law factors.) The independence does not extend to the draw values themselves: an observer of which card went to which bin reconstructs the permutation entirely, which is why observation of the draws equals compromise.*

Equivalently, the procedure sorts the cards by infinite random d -ary strings—by random reals—revealing only as many digits as needed to break ties; and it is a physical realization of the strong uniform time of Aldous and Diaconis [1] for riffle-type shuffles: deal by random digits until all cards’ strings are distinct, at which instant the deck is exactly uniform. Truncating the recursion at a fixed number of rows, ties broken by deal order, recovers the classical riffle-shuffle family of Bayer and Diaconis [2] with its small but nonzero deficits; running the recursion to completion removes the deficit entirely, which is why no distance tables, entropy accounting, or pass-count rules appear in this paper. Zero requires no bookkeeping.

3 The one discipline

The hypothesis that collection order is fixed *before* the deal is load-bearing, and the natural human behavior violates it: pile sizes are visible, and a tidy operator reaches for the largest pile first. This biases the result. Already at $n = 3$ cards and $d = 2$, fixed collection realizes

five of the six permutations while largest-pile-first realizes only three, with total-variation error $1/2$; the output would always begin with its longest run of previously-adjacent cards. Independence from the draws is the true requirement, and predetermination—the same order every row, agreed before the first roll—is its enforceable form. In multi-party use this is a security requirement, not hygiene: whoever collects the piles otherwise holds a channel for biasing the deck while appearing merely tidy.

Computational form. Although stated as a manual scheme, the rule is directly a computer algorithm, and at $d = 2$ it needs nothing but a fair-bit generator: stably partition the array by one fresh random bit per element, recurse on every block of two or more, concatenate. The output is exactly uniform by the theorem—no modular reduction, no range-rejection loop, no floating point—and by the corollary its running time is provably independent of the output, a timing-side-channel guarantee that implementations of Fisher–Yates must argue for separately. The expected cost is $n \log_2 n + O(n)$ bits (simulated: 365 bits at $n = 52$, 266 at $n = 40$, against the entropy lower bound $\log_2 n!$ of 226 and 159; overhead factor about 1.6). The computational trick itself is folklore—sorting by lazily generated random reals—but it inherits, unchanged, the exactness proof and the transcript-independence corollary above.

4 Multi-party use

When N parties require a deck none can bias, each rolls its own randomizer for every card and the bin is the sum modulo d of the parties’ draws (per die; for coins, the XOR of flips). The combined draw is exactly uniform provided at least one party draws fairly and independently, so a dishonest coalition of $N - 1$ loaded-dice players gains nothing. Draws for a card are thrown together or pre-committed, so loaded hardware cannot adapt to honest outcomes; collection needs no trust, being predetermined and observable; and by the theorem the procedure may be—should be—conducted with every entitled party watching everything.

5 The apparatus

The radix need not be constant across levels—the theorem’s hypotheses are per-level—and small piles need not recurse at all when their symmetric group can be sampled directly. Both freedoms condense the procedure into a purpose-built stand, cuttable on a laser cutter: **36 labeled bins** tipped at 60° for the first deal (addressed by two dice, row and column); **two rows of six bins** at the right for recursion (one die); and, along the bottom, **an action table** for the die faces: 1, 1X, 2, 2X, 3, 3X.

The operator deals the deck into the 36 bins, then sweeps them in the stand’s manufactured order. A bin of one card goes to the output. Small piles resolve by one roll against the action table, which labels permutations by *actions* rather than patterns: the digit names which card (1 = top) moves to the back, and X says whether to exchange the remaining pair. For a three-card bin this factors S_3 as $3 \times 2 = 6$, one action per die face, exact; for a two-card bin only the X matters—even faces exchange—exact, three faces each way. There is no permutation decoding, only two motions: pull a card to the back, swap or don’t. A bin of four resolves in three throws, deterministically, because 4 divides 36: the two dice name a cell, and each cell of the stand carries a small corner numeral—its index mod 4—so the rig

does the arithmetic; the numeral picks the first card of the pile (exactly uniform, nine cells per residue), and one further die applies a three-card pattern to the rest. Only a bin of five or more cards (about one per two 52-card shuffles) is dealt into a six-bin row and resolved the same way; two rows are ample (cleared rows are reused), and a majority of 52-card runs never leave the first level at all. The whole shuffle is thus, in expectation, one deal of the deck plus roughly a dozen and a half single-roll fix-ups.

Two remarks. First, the stand *manufactures the discipline* of Section 3: the collection order and the permutation table are cut and labeled into the artifact, so the theorem’s one behavioral hypothesis is enforced by geometry rather than operator virtue—for a key-generation apparatus, a property worth having. Second, the action labels exist for the same reason: pattern strings (“201” vs. “210”) invite systematic misreads, and a systematic misread skews the six-way distribution; “move card 2 to the back, exchange” does not. What reading error remains is value-independent noise—the operator sees no faces. A humbler rig bootstraps the same procedure from four identical slotted shoebox lids: three tiled side by side form the 36-bin field (two dice: column and row), and the fourth carries the two recursion rows and the printed enumeration. A one-page cut template (print four copies at full scale) accompanies the paper; a single such lid, addressed by a die and a coin, is by itself the twelve-slot rig of the cost table.

6 Cost

Simulated (placements equal draws for the plain recursion; the stand and shoebox rows count randomizer throws, their single-roll fix-ups included; “rows” is the recursion depth reached):

	placements		rows	
	mean	95th pct.	mean	95th pct.
$n = 52, d = 6$ (die)	157	168	4.8	6
$n = 52, d = 8$ (coins)	138	148	4.3	5
$n = 52, d = 12$ (die+coin)	121	129	3.6	5
$n = 52$, shoebox (12 slots, 2/3-fix)	105	110	2.8	3
$n = 52$, stand (36, $d6$ rows, 4-fix)	73	79	1.5	2
$n = 40, d = 12$	89	96	3.4	4
$n = 40$, stand	53	58	1.2	2

The recursion re-deals only collisions, never the whole deck, so the work concentrates tightly around its mean; on the stand, the level-one deal (52 throws, 52 placements) is in expectation nearly the entire shuffle—expected placements total 55, and a majority of runs never recurse. For comparison, achieving merely 10^{-3} total-variation error with fixed full-deck passes costs 156 placements at $n = 52$; the exact procedure with die-and-coin costs less and errs by nothing.

7 Cautions

The randomizers are the entire entropy source; the theorem is conditional on fair, independent draws—use quality dice or the multi-party rule. A misdealt card (wrong bin) is harmless

outcome-independent noise, but a collection order improvised after seeing the piles is not (Section 3). Anyone who observes the draws can reconstruct the deck: for a secret key, admit exactly the parties entitled to it, and never reuse a recorded sequence. Simulation code reproducing the cost table and an empirical uniformity check accompanies the paper (`verify_shuffle.py`, commands `recursive` and `stand`).

References

- [1] D. Aldous and P. Diaconis, “Shuffling cards and stopping times,” *American Mathematical Monthly* **93** (1986), 333–348.
- [2] D. Bayer and P. Diaconis, “Trailing the dovetail shuffle to its lair,” *Annals of Applied Probability* **2** (1992), 294–313.
- [3] B. Schneier, “The Solitaire encryption algorithm,” 1999, <https://www.schneier.com/academic/solitaire/>.